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Microtia: Epidemiology & Genetics

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Abstract

Microtia is a congenital anomaly of the ear that ranges in severity from mild structural abnormalities to complete absence of the ear, and can occur as an isolated birth defect or as part of a spectrum of anomalies or a syndrome. Microtia is often associated with hearing loss and patients typically require treatment for hearing impairment and surgical ear reconstruction. The reported prevalence varies among regions, from 0.83 to 17.4 per 10,000 births and the prevalence is considered to be higher in Hispanics, Asians, Native Americans, and Andeans. The etiology of microtia and the cause of this wide variability in prevalence are poorly understood. Strong evidence supports the role of environmental and genetic causes for microtia. Although some studies have identified candidate genetic variants for microtia, no causal genetic mutation has been confirmed. The application of novel strategies in developmental biology and genetics has facilitated elucidation of mechanisms controlling craniofacial development. In this paper we review current knowledge of the epidemiology and genetics of microtia, including potential candidate genes supported by evidence from human syndromes and animal models. We also discuss the possible etiopathogenesis in light of the hypotheses formulated to date: neural crest cells disturbance, vascular disruption and altitude.

Keywords

microtia; anotia; craniofacial development; craniofacial microsomia; hemifacial microsomia; OAVS (oculo-auriculo-vertebral spectrum)

INTRODUCTION

The vertebrate ear is divided into the outer, middle and inner ear. This review focuses on malformations of the external ear, and more specifically of the auricles, which are collectively termed microtia. However, other components of the external (acoustic meatus and tympanic membrane), middle, and inner ear are also frequently impacted, as are other craniofacial and extra cranial anomalies.

Microtia encompasses a spectrum of congenital anomalies of the auricle that range in severity from mild structural abnormalities to complete absence of the ear (anotia) [Carey et al., 2006]. There is no consensus regarding the terminology used for these external ear malformations. Some authors prefer to use the term “microtia” [Alasti and Van Camp, 2009; Castilla and Orioli, 1986; Hunter et al., 2009a; Suutarla et al., 2007] while others use “microtia-anotia” or “microtia/anotia” [Canfield et al., 2009; Forrester and Merz, 2005;

Harris et al., 1996; Mastroiacovo et al., 1995; Shaw et al., 2004; Stevenson, 2006]. In this paper, the term “microtia” includes anotia as the most severe end of the microtia spectrum.

The occurrence of microtia is of public health importance in part due to the psychosocial sequelae, including the stigma associated with malformations of the ear and the burden of undergoing multiple surgeries [Du et al., 2007; Li et al., 2010; Steffen et al., 2010]. In addition, greater than 90% of individuals with microtia experience conductive hearing loss on the affected side [Bassila and Goldberg, 1989; Calzolari et al., 1999; Carey et al., 2006; Ishimoto et al., 2007; Suutarla et al., 2007]. Although there has been no recent review of the average medical cost associated with treatment of microtia and the associated health issues, the costs are expected to be considerable given that hearing impairment care and multiple surgical procedures for ear reconstruction are often necessary.

Microtia occurs more frequently in males, with an estimated 20–40% increased risk compared to females. Microtia can occur bilaterally, although 77–93% of affected individuals have unilateral involvement [Canfield et al., 2009; Castilla and Orioli, 1986; Forrester and Merz, 2005; Mastroiacovo et al., 1995; Nelson and Berry, 1984; Shaw et al., 2004; Suutarla et al., 2007]. The right ear is affected in approximately 60% of individuals with unilateral microtia [Castilla and Orioli, 1986; Forrester and Merz, 2005; González-Andrade et al., 2010; Harris et al., 1996; Mastroiacovo et al., 1995; Suutarla et al., 2007]. A higher proportion of bilateral microtia is found in cases with associated anomalies not directly related to the ear abnormality [Canfield et al., 2009; Harris et al., 1996; Mastroiacovo et al., 1995; Shaw et al., 2004]. Microtia may occur as an isolated condition, or as part of a spectrum of anomalies or a syndrome. The most common anomalies associated with microtia include: vertebral anomalies, macrostomia, oral clefts, facial asymmetry, renal abnormalities, cardiac defects, microphthalmia, holoprosencephaly, and polydactyly [Carey et al., 2006; Harris et al., 1996; Kaye et al., 1989; Mastroiacovo et al., 1995]. Most of these anomalies are also associated with oculo-auriculo-vertebral spectrum (OAVS), a condition notable for wide clinical variability and for which the etiologies remain unknown.

Existing data indicate that Mendelian inheritance is more likely in syndromic and familial cases of microtia, whereas multifactorial or polygenic causes are more probable in sporadic cases. Several non-genetic factors have been consistently associated with microtia. Although no genes have been associated with isolated microtia, a number of genes have now been identified on syndromes associated with microtia. The purpose of this paper is to review the current literature regarding the genetics and epidemiology of microtia, and discuss the etiological and pathogenetic mechanisms proposed for this condition.

Prevalence

Population-based studies on microtia prevalence conducted in Italy, France, Sweden, Finland and United States show prevalence rates ranging between 0.83 and 4.34 per 10,000 births [Canfield et al., 2009; Forrester and Merz, 2005; Harris et al., 1996; Shaw et al., 2004; Suutarla et al., 2007] (Table I). This wide range in prevalence may be due to variation among the studies in case inclusion criteria and case ascertainment. Microtia is an external anomaly that can be identified on physical examination of newborns; however, the less severe forms of microtia may not be recognized or described as a form of microtia in medical records or the term may be used for deformities of the ears. This could lead to under or over reporting of microtia in reports of prevalence.

Population-based studies performed in the United States however, consistently report variations in prevalence according to race/ethnicity, with a higher risk for individuals of Asian heritage [Forrester and Merz, 2005; Harris et al., 1996; Shaw et al., 2004], Pacific

Islanders [Forrester and Merz, 2005] and individuals of Hispanic descent [Canfield et al., 2009; Harris et al., 1996; Shaw et al., 2004; Yang et al., 2004] when compared to Caucasians and African Americans. Studies conducted using non population-based data reported higher prevalence for Ecuadorians [Castilla and Orioli, 1986; González-Andrade et al., 2010], Chileans, and among Native Americans in the US [Aase and Tegtmeier, 1977; Jaffe, 1969; Nelson and Berry, 1984]. More comprehensive studies are required to investigate the racial/ethnic differences in prevalence of microtia and the etiology of this variability. For instance, the prevalence of microtia is three to eight times higher in Chile and Ecuador than previously reported worldwide, which may be at least in part due to genetic variation, environmental factors (such as diet) or a combination of gene-environment interactions.

Human disorders with microtia

Approximately 20-60% of children with microtia have associated anomalies or an identifiable syndrome (Table II) [Castilla and Orioli, 1986; Kaye et al., 1989; Mastroiacovo et al., 1995; Shaw et al., 2004], therefore individuals with microtia should be examined for other dysmorphic features. Microtia is a common feature of craniofacial microsomia, Townes-Brocks syndrome and the mandibulofacial dysostoses (e.g.: Treacher-Collins and Nager syndrome) and these conditions should be considered among the differential diagnosis when evaluating an individual with microtia.

Oculo-auriculo-vertebral spectrum (OAVS), is characterized by facial asymmetry, microtia, ear and facial tags, epibulbar dermoids, microphthalmia, and macrostomia. [Heike and Hing, 2009]. Craniofacial, or hemifacial, microsomia and Goldenhar syndrome are included in this spectrum. Extracranial features include renal, cardiac and vertebral anomalies. There is no agreement about minimal diagnostic criteria for OAVS. Most cases of OAVS are sporadic, however autosomal dominant or, less commonly, recessive inheritance have been reported.

Microtia and OAVS share the following characteristics: 1) variable phenotypic expression, 2) asymmetric involvement of facial structures, 3) right side preponderance, 4) male predilection, and 5) familial occurrence of microtia or related anomalies such as preauricular tags and pits. Based on these observations, it has been suggested that isolated microtia represents a milder phenotype of OAVS [Llano-Rivas et al., 1999; Rollnick and Kaye, 1983; Tasse et al., 2005]. This has led to the controversial concept that most (or all) cases presenting with apparent isolated microtia are actually cases of OAVS. This controversy remains unsettled. In many cases, the occurrence of microtia associated with chromosomal abnormalities and in single gene disorders supports a complex genetic regulatory network coordinating morphogenesis of the external ear. Therefore, although the clinical expression of microtia and OAVS overlap and likely share many common genetic mechanisms, each should be considered as a separate entity. In this review, we cite the literature referring to microtia as a separate condition from OAVS.

Known Risk factors

To date there have been few published case-control studies on microtia [Castilla and Orioli, 1986; Correa et al., 2008; Ma et al., 2010; Mastroiacovo et al., 1995; Zhang et al., 2009]. The risk factors that were identified in these studies include low birth weight, higher maternal parity, maternal acute illness and use of medications (specific acute maternal conditions or medications were not identified in these studies), and maternal diabetes mellitus. Multiple births, advanced maternal age, low maternal education and Hispanic ethnicity have also been reported as risk factors for microtia in cross-sectional, population-based studies. More recently, periconceptional intake of folic-acid-containing supplements has been associated with reduced risk of microtia among non-obese women [Ma et al.,

2010]. A summary of the risk factors reported in the literature, in case-control and cross-sectional studies, is presented in Table III.

Strong evidence supports the association between gestational exposure to specific medications and microtia, including well known teratogens such as retinoids, thalidomide and the immunosuppressant, mycophenolate mofetil [Anderka et al., 2009; Klieger-Grossmann et al., 2010]. Alcohol has been inconsistently reported as a risk factor [Carey et al., 2006]. The mechanisms by which these exposures cause microtia have not been fully elucidated.

High altitude, usually defined as above 2,500 meters or 8,200 feet, has been associated with microtia in two independent studies in South America [Castilla et al., 1999; González-Andrade et al., 2010], which is inhabited by the largest populations living at high altitudes in the world. The observed association may be related to altitude or altitude could be a confounder. For example, the true association may be related to ethnicity, given the high proportion of Native American ancestry in regions of high altitude, or to differences in diet between low and high altitude populations.

Overview of Classification systems

The degree of phenotypic variability of congenital anomalies of the ear makes the development of a meaningful classification system challenging (Figure 1). Nevertheless, classification systems can facilitate diagnosis, treatment and standardized data collection in multi-center studies. Hermann Marx published the first system, named the Marx classification, in 1926 and it remains one of the most frequently used systems [Marx, 1926]. Tanzer classified ear abnormalities correlating with the surgical approach [Tanzer, 1978]. Weerda et al. [1988] modified the Marx and Tanzer definitions based on embryologic development as well as surgical steps and included all congenital abnormalities of the external ear (i.e., deformities and minor anomalies). The American Journal of Medical Genetics has recently published a collection of articles in an effort to standardize external ear terminology in the clinical genetics field [Hunter et al., 2009a]. The Weerda classification system was chosen as the basis for the standardized terminology used for microtia. These classification systems, commonly cited in studies of microtia, are summarized in Table IV.

Most published studies on microtia report the presence or absence of microtia and/or anotia without further detail regarding severity. This is likely due, in part, to the fact that many prevalence studies of microtia rely on birth defect registries, which incorporate the International Classification for Diseases (ICD) coding system. The ICD system has only one code for microtia and one code for anotia and no information on severity or laterality.

As the genetic control of embryonic tissue morphogenesis is better understood, we may discover that the existing classification systems are too simplistic to be used in the study of normal and abnormal ear development. Detailed description of the malformation of each component of the ear, and acquisition of corresponding images should be the standard for recording information on microtia and other birth defects, regardless of the classification system chosen. Epidemiological and genetic studies could benefit from more detailed phenotypic information that would enable subclassification and grouping of malformations with shared characteristics. As we develop new genomic approaches for the study of birth defects, the importance of detailed phenotypic description has become clear. For example, in the discovery of the causative gene for Kabuki syndrome, reassessment of the images and clinical description was crucial when the first attempt of exome sequencing was unsuccessful [Ng et al., 2010]. Likewise, subphenotype analysis indicates that at least a subgroup of isolated cleft lip may be etiologically distinct from isolated cleft lip and palate

[Jugessur et al., 2011]. Detailed description of external ear malformations would enable future reassessment of this information and re-classification if necessary to aggregate cases in multiple ways. The feasibility of this type of approach has been demonstrated in a study that performed systematic examination of ears of individuals with Cornelia de Lange syndrome and controls, using standardized 2D photographs [Hunter et al., 2009b].

EMBRYOLOGY: OUTER EAR DEVELOPMENT

Vertebrate embryos develop a series of paired outgrowths on the ventro-lateral surface at their rostral end called the pharyngeal or branchial arches, which give rise to structures of the head and neck [Schoenwolf and Larsen, 2009b]. The pharyngeal arches are composed of mesenchymal cells of mesodermal and cranial neural crest origin. The neural crest cells (NCC) are a transitory group of pluripotent cells that originate from the dorsal part of the embryonic neural tube: the ectodermal-neurectodermal boundary. During early development, many of these cells collectively transform to a mesenchymal phenotype and assume new morphological characteristics distinct from their epithelial neighbors, segregate from the neural tube and emigrate through specific routes to contribute to a wide variety of tissues and structures throughout the vertebrate body [Engleka et al., 2008]. In the cranial region, reciprocal signaling between neural crest cells (ectomesenchyme) and other embryonic cell types (e.g. endothelia and craniofacial ectoderm) play an important role in driving facial outgrowth and morphogenesis, including that of the external ear [Noden and Trainor, 2005].

The outer ear consists of the ear pinna (i.e. auricle, external ear), the external acoustic meatus (i.e. ear canal), and the outer layer of the tympanic membrane (i.e. eardrum). Outer ear development is driven by the mesenchyme of the first and second pharyngeal arches and is controlled, at least in part, by genes that determine first and second pharyngeal arch identity.

The auricle is formed from several protuberances in the first and second arches known as auricular hillocks (i.e. hillocks of His). These hillocks surround the first pharyngeal cleft, which is the space between the first and second arches. Each of the hillocks contributes to a specific component of the pinna, and those in the second arch form most of the ear structure [Mallo, 2003]. The auricular hillocks grow, fuse and undergo morphogenesis to produce an appendage that funnels airborne vibrations into the meatus and along the canal to the tympanic membrane.

The outer ear begins its development during the fifth week, and the hillocks are first identifiable during the sixth week of embryogenesis. The development of the auricular hillocks into an auricle progresses slowly over several months and takes place largely during fetal stages. From their initially low position on the embryonic neck the auricles re-position progressively dorsalward [Schoenwolf and Larsen, 2009a]. As with more general facial growth [Hu and Marcucio, 2009], the overlying pharyngeal ectoderm may play a key role in determining the overall morphology or form of the auricle.

The auditory canal and tympanic membrane are derived from ectoderm of the pharyngeal cleft that separates the first and second pharyngeal arches. The cleft invaginates to form the meatus; this process is controlled and coordinated by a C-shape skeletal structure, the tympanic ring, which develops from the first arch mesenchyme. As the ring grows, the invaginated external acoustic meatus starts to flatten down in the plane defined by the ring and becomes apposed to the endoderm of the middle ear cavity [Mallo, 2003]. The ring progressively integrates into the temporal bone at postnatal stages to serve as the attachment of the tympanic membrane.

Auricular and external acoustic meatus development must be tightly coordinated in order to be functional. Evidence from two mouse models (*Gsc* and *Prx1* mutant lines), however, suggests that auricular and external acoustic meatus development is regulated by independent mechanisms, as both the *Gsc* and *Prx1* mutants present with absent external acoustic meatus but exhibit fairly normal auricles [Martin et al., 1995; Yamada et al., 1995].

GENETICS OF MICROTIA

Investigators have used a variety of genetic approaches to study microtia, including linkage analysis, direct sequencing of DNA from affected individuals, the study of single gene disorders that occur with microtia, identification of cytogenetic rearrangements in cases, and the study of animal models.

Evidence for a significant genetic contribution to microtia is based on: 1) higher concordance in monozygotic twins than in dizygotic twins; 38.5% and 4.5%, respectively [Artunduaga et al., 2009]; 2) reported familial cases with autosomal recessive or dominant modes of inheritance with variable expression and incomplete penetrance [Alasti et al., 2008; Balci, 1974; Balci et al., 2001; Chafai Elalaoui et al., 2010; Ellwood et al., 1968; Guizar-Vazquez et al., 1978; Gupta and Patton, 1995; Klockars et al., 2007; Konigsmark et al., 1972; Orstavik et al., 1990; Schmid et al., 1985; Strisciuglio et al., 1986; Zankl and Zang, 1979]; 3) estimates of familial cases ranging from 3 to 34% [Castilla and Orioli, 1986; Llano-Rivas et al., 1999; Mastroiacovo et al., 1995; Okajima et al., 1996]; 4) more than 18 different microtia-associated syndromes for which single-gene defects or chromosomal aberrations have been reported; and 5) mouse models demonstrating that mutations in specific genes result in microtia. We discuss below the most relevant existing data for candidate genes from studies on animal models and humans.

Animal models studies

The application of novel strategies (analytical, genetic, imaging, etc) in developmental biology and genetics has begun to facilitate elucidation of mechanisms controlling craniofacial development in animal models. Murine models in particular are commonly used to study developmental mechanisms involved in the formation of the head and face. Defects in outer ear development in mutant mice range from hypomorphisms to the complete absence of structural elements. In Table V, we have included models with abnormalities in the structure of the external ear to be consistent with the definition of microtia. Mouse models with only inner ear anomalies were not included. In this section, we discuss in further detail some of the most promising models.

Hox Genes—Homeobox genes are involved in the development of the pharyngeal arches. They encode highly conserved transcription factors that control positional identity of cells (body patterning) and morphogenesis throughout development, as well as switch on cascades of other genes. The *Hox* gene family is clustered within the genome and is ordered on the chromosome in the sequence in which they are expressed during development; this highly ordered pattern of gene expression might constitute part of a mechanism whereby morphogenetic specification is established [Kmita and Duboule, 2003]. Inactivation of *Hoxa1* in mice results in hypoplastic external ears and abnormalities of the middle and inner ear, whereas compound *Hoxa1/Hoxb1* mutants present with complete anotia [Gavalas et al., 1998]. In contrast, *Hoxa2* seems to be required for defining second pharyngeal arch identity and thus the initial steps of pinna formation, and is strongly expressed in the pinna of mice. Consistent with this, *Hoxa2* knockout mice present with microtia, described as “a small protuberance with no recognizable shape” [Gendronmaguire et al., 1993]. *Hoxb6* and *Hoxa7* deficient mice present with microtia in addition to open-eyes and cleft palate [Balling et al., 1989; Kaur et al., 1992].

Six and Eya Genes—In vertebrates, members of the *SIX* homeobox gene family (*SIX1–6*) have also been implicated in external ear development [Kawakami et al., 2000]. *SIX* genes are homologs of *sine oculis (six)* gene in the vinegar fly, *Drosophila melanogaster*. *SIX* function seems conserved across evolution since knock-down of *Six1* in frogs, chicks, and mice result in craniofacial abnormalities [Brugmann et al., 2004; Christophorou et al., 2009; Laclef et al., 2003] while misexpression of *Six2* leads to frontonasal dysplasia in mice [Fogelgren et al., 2008]. *Six1/Six4* mice present with microtia, whereas *Six1* deficiency alone is associated with normal external and middle ears [Laclef et al., 2003], suggesting some redundancy in function within this gene family. Other *Six* mutants have not been reported to have ear abnormalities.

EYA1 is the human homolog of the *Drosophila eyes absent (eya)* gene. *EYA* forms a complex with *SIX* (*EYA-SIX*) to regulate the development of several tissues and organs in vertebrates and in flies. Natural target genes of the *EYA-SIX* complex include *SIX2* and *SALL1*. Studies on *Eya1* expression have shown a major role in pinna development apparently related to cartilage formation; the knockout mice for *Eya1* present anotia. *Sall1* is expressed in craniofacial tissue but the knockout animals have normal ears. As for the *SIX* genes, there are additional *SALL* genes in mammals (*SALL2–4*) and so redundancy in function may also mask or modulate the phenotypic presentation.

Recently, *Sip1* (Shank-interacting protein-like 1) and *Rbck1* (RBCC protein interacting with PKC1) were identified as novel *Eya1*-interacting proteins. Both *Sip1* and *Rbck1* are expressed together with *Eya1* in many tissues in mouse and zebrafish to direct the development of the inner ear and the pharyngeal arches as well as other organs. In fact, both *Sip1* and *Rbck1* act as cofactors for the *Eya-Six* complex [Landgraf et al., 2010]. Further experiments regarding the functional consequences of the interaction of *Sip1* or *Rbck1* with *Eya1* should clarify the importance of the respective interaction for outer ear development in mammals.

Tbx1—In mice, mutations in *Tbx1*, a member of the T-box gene family of transcription factors, result in failure of middle and outer ear development and in hypoplasia of the inner ear sensory organs. A similar phenotype was also seen following inactivation of *Tbx1* exclusively in pharyngeal arch endoderm, indicating a primary role for this gene in pharyngeal arch morphogenesis [Arnold et al., 2006]. Of interest, *Tbx1* heterozygosity is associated with chronic otitis media, but not morphological defects, and does not interfere with the formation of the outer, middle and inner ear structures [Liao et al., 2004].

Irf6 and Chuk (Ikka)—Mice homozygous null for *Irf6* lack external ears in addition to exhibiting abnormal skin, limbs and both shorter snouts and jaws. Ectopic epidermal adhesions at several sites, including the oral cavity, between the tail and hindlimbs and in the esophagus were observed, although not specifically reported for the ear. A similar phenotype was observed in mice deficient for *Chuk* (also known as *Ikka*). The authors speculate, based on histological and gene expression analyses, that the abnormalities in the *Irf6* and *Chuk* mice are secondary to defects in epidermal differentiation or cell proliferation [Hu et al., 1999; Ingraham et al., 2006].

Signaling pathways—Signaling pathways involved in the outer ear development include bone morphogenetic proteins (Bmps), Wingless/INT (Wnts), fibroblast growth factors (Fgfs), and retinoic acid. Dysregulation of these signaling pathways triggered by genetic or environmental factors constitutes a potential source of under- or maldevelopment. While NCC likely receive patterning signals during migration, much of the signaling necessary for patterning within an arch comes from signals received after their arrival at the arches [Knight and Schilling, 2006].

The Bmp genes, especially *Bmp5*, have been considered as candidate genes for microtia in humans; however, studies in mice have shown that Bmp5 is apparently more related to growth than the early pattern of differentiation and formation of the external ear. The *Bmp5* mutant mice usually present with short ears attributed to defective auricular cartilage framework. Over two dozen viable radiation- and chemically-induced alleles have been isolated at the Bmp5 locus (Russell 1971; Russell et al. 1989; Kingsley et al. 1992; Marker et al. 1997). The different mutations produce an apparent gradient of effects on the size of the external ear; mutants completely missing the *Bmp5* gene have the shortest ears.

FGF signaling, involving different Fgf ligands and their receptors, Fgfr1-3, plays various roles in pinna development [Abu-Issa et al., 2002; Wright and Mansour, 2003] as evidenced by specific mutant phenotypes; *Fgf8* and *Fgf10* mutant mice present with small outer ears [Abu-Issa et al., 2002; <http://www.informatics.jax.org>], and mice homozygous for a hypomorphic *Fgfr1* allele present with very small ears and abnormal external auditory canals [Partanen et al., 1998]. However, it is not clear when these signaling components are required, nor whether these particular ligands and receptors are expressed in the pinna during late gestation.

Members of the Wnt family have been implicated in NCC formation and development, but their independent roles have been difficult to determine due to overlapping expression and functional redundancy. It has been shown that Wnt5a is expressed in the mesenchyme of the developing outer ear, and indeed *Wnt5a* knockout mice present with small ears [Yamaguchi et al., 1999]. However, microtia has not been described in any other Wnt mutant.

Mouse lines harboring a mutation in endothelin or endothelin receptors also present with various ear malformations. The endothelin pathway has a well-established role in regulating neural crest proliferation and migration, and therefore it is plausible that mutations in this pathway could be involved in microtia in humans. In this regard, the transcription factor *Gooseoid* (*Gsc*), a downstream target of endothelin signaling, is expressed in the pharyngeal mesenchyme around the first pharyngeal cleft and has been implicated in outer and middle ear development through mutational analyses in patients (see below).

Human genetics studies

Microtia has been reported in individuals with autosomal trisomies, such as trisomy 18 (Figure 2), 21, and 22, as well as with mosaicism of trisomy 13 and 18 [Giannatou et al., 2009; Griffith et al., 2009]; and aneusomies, as in deletion of 4p, 5p and 18p, 18q, and 22q11.2. Chromosomal translocations involving the 6p24 region have been associated with orofacial clefting and bilateral microtia [Davies et al., 1998]. Several cases reports of mosaicism 46,X,der(Y)t(Y;1)(q12;q21)/46,XY describe the presence of microtia associated with anomalies such as kyphoscoliosis, oligodactyly, joint contractures, central nervous system malformations, omphalocele, diaphragmatic hernia, cardiac defects, and urogenital malformation [Li, 2010; Scheuerle et al., 2005; Watson et al., 1990; Zeng et al., 2003]. Microtia has been associated with abnormalities in each of the chromosomes [POSSUM: A dysmorphology database of multiple malformations, metabolic, teratogenic, chromosomal and skeletal syndromes and their images - for learning diagnosis, 2010] confirming Schinzel's observation that malformations confined to one or very few chromosome aberrations are suspicious for single gene deletions, whereas, malformations frequent in chromosome aberrations are caused by deficiency of a step in organogenesis [Schinzel, 2001]. For the purpose of this review we have only cited cytogenetic rearrangements recurrently reported involving microtia.

Microtia is a clinical finding in several well established human single gene disorders. For example, mutations in *SIX1* and *EYA1* have been shown to cause Branchio-otic (BO)

syndrome, while mutations in *SIX5* and *EYA1* can cause Branchio-oto renal (BOR) syndrome, both are associated with microtia, among other craniofacial defects [Abdelhak et al., 1997; Hoskins et al., 2007; Kumar et al., 1997; Rodriguez Soriano, 2003]. Other familial cases with syndromic microtia have also been reported. Table II summarizes the human genes involved in syndromes that are associated with microtia.

Few studies have focused on the genetic causes of isolated microtia. Sequence analysis of *GSC* exons in 121 individuals with isolated microtia revealed a missense mutation in exon 3 in two cases. In the same study, screening of the *BMP5* locus revealed a missense mutation in four patients. None of these mutations were detected in control subjects, suggesting a causative role. Individuals with incomplete clinical data, inadequate quantity of blood samples, or unsatisfactory genetic analyses were excluded and thus the total number of cases and controls included in the analysis is not clear [Zhang et al., 2010].

Monks et al did not identify mutations in exons of *HOXA2* or *SIX2*, which acts downstream of *HOXA2* during development, in 12 patients of Hispanic and African descent with isolated microtia [Monks et al., 2010]. In another study, the methylation status of the *EYA1* gene promoter was analyzed in 64 individuals with microtia and 36 healthy controls. The authors reported that the methylation levels at this locus were significantly lower in individuals with microtia than in controls and suggested that hypomethylation may be related to the pathogenesis of this condition [Lin et al., 2009]. Further studies are needed to validate these conclusions.

In summary, although some studies have found genetic variants potentially associated with microtia, no causal genetic mutation has been confirmed to date.

CURRENT HYPOTHESES FOR THE ETIOLOGY AND PATHOGENESIS

Microtia is both etiologically and pathogenetically heterogeneous. As discussed above, single gene mutations are associated with microtia in syndromic and familial cases, whereas a multifactorial (genetic and environmental) or polygenic cause is likely in sporadic cases. Current hypotheses favor disturbance of NCC as the likely underlying cause, although the exact mechanism(s) remain unknown. However, given the clinical heterogeneity, it is possible that different pathogenetic processes lead to the different types of microtia. We discuss several hypotheses for the occurrence of this condition below.

Neural crest cells disturbance

Defects in NCC function have been associated with numerous craniofacial syndromes [Passos-Bueno et al., 2009]. In Treacher Collins syndrome, *TCOF1* mutations result in haploinsufficiency of the protein Treacle (encoded by *TCOF1*) leading to insufficient ribosome genesis, diminished cell proliferation, and increased neuroepithelial apoptosis. The mechanism proposed is that this results in depletion of NCC precursors with reduced number of cells migrating into the first and second pharyngeal arches leading to the complete craniofacial phenotype observed in the syndrome that includes severe, bilateral microtia [Trainor, 2010].

Strong evidence for the role of NCC in the occurrence of microtia derives also from the recent studies on the causative mechanisms of various teratogens associated with this condition. Retinoid and diabetic embryopathy have been associated with apoptosis of NCC before migration into the pharyngeal arches, and disturbance of NCC differentiation after arrival in the pharyngeal arches. In diabetes, hyperglycemia has been recently associated with down regulation of *Pax3*, which encodes a transcription factor critical for early NCC survival and migration [Zabihi and Loeken, 2010]. Conversely, retinoid exposure appears to

disrupt the endothelin signaling pathway; which in turn regulates *Hox* gene expression. *Hox* genes are hypothesized to govern positional identity of NCC before and during migration from the neural folds. Mallo [2003] suggested that the more severe forms of microtia could result from a loss of second arch identity, since most of the definitive pinna derives from the hillocks of this arch. This conclusion is supported for the phenotype observed on the *Hoxa2* mouse mutant (anotia), a gene that is strongly expressed in the second pharyngeal arch. The effects of thalidomide could include down regulation of *Fgf8* [Hansen et al., 2002] and *Bmp* signaling [Ito et al., 2010; Knobloch et al., 2007], though direct anti-angiogenic effects and oxidative stress are also postulated as independent mechanisms [Ito et al., 2010; Parman et al., 1999].

From an embryological and developmental biology perspective, defects or insults affecting NCC delamination, proliferation, apoptosis or migration, or their reciprocal interactions with mesoderm, endoderm, or overlying ectoderm are feasible explanations for the impairment in auricular hillock growth, re-positioning, or cartilage development seen in patients with various forms of microtia. In view of this, investigations into the pathogenesis of 22q11.2 deletion syndrome (22q11.2DS) initially focused on intrinsic abnormalities in NCC. *TBX1* is a gene located in the typically deleted region in this condition and is considered to be a candidate gene for several of the malformations associated with 22q11.2DS. *TBX1* encodes a transcription factor, which presumably affects the expression of a secreted or cell surface molecule; however, *TBX1* appears not to be expressed by the NCC, but to be expressed throughout the non-crest pharyngeal mesenchyme and in regions of pharyngeal endoderm. Individuals with 22q11.2DS often have small ears and there are reports of some presenting with microtia [Digilio et al., 2009]; *Tbx1* mice mutants also can present with small or absent ears [Liao et al., 2004]. Thus, an indirect interaction between non-crest mesoderm or endoderm and neural crest can alter NCC fate and result in craniofacial malformations that might include microtia.

Vascular disruption

Vascular disruption can occur via several mechanisms, including (1) occlusion of an artery that interrupts blood flow to previously formed tissue, (2) vasoconstriction and diminished arterial blood flow, or (3) underdevelopment of the arterial system required for adequate blood supply to developing tissues. Vascular disruption has been proposed to cause microtia by disruption in the development of the blood vascular system in the head and neck, resulting in localized ischemia and tissue necrosis, although this is still heavily debated [Sadler and Rasmussen, 2010].

The concept of vascular involvement in microtia comes from various observations. The greater prevalence of unilateral cases of microtia suggests a more localized effect during embryogenesis, which could feasibly result in occlusion of a single vessel. This hypothesis is mainly supported by studies in the 1970s [Poswillo, 1973; Poswillo, 1975] in which mice exposed to triazine and monkeys exposed to thalidomide showed ipsilateral hematomas at the junction of the pharyngeal and hyoid arteries with associated unilateral ear and mandibular defects. Additionally, Otani et al [1991] and Naora et al [1994] reported a phenotype resembling craniofacial microsomia (unilateral microtia, abnormal biting, anomalies in the EAM and middle ear, and cranial base) in a transgenic mouse line carrying a non-expressed transgene. Notably, the authors reported rupture of the vasculature of the second pharyngeal arch with histologically confirmed hemorrhage and subsequent phagocytosis. They concluded that integration of the transgene on mouse chromosome 10 interrupted an endogenous gene that has a critical role in craniofacial morphogenesis. The causative gene in this important mouse model has not been identified.

Arguing against a primary role for vascular disruption, Johnston and Bronsky [1995] re-assessed Poswillo's original experiments and concluded that the hematomas appeared too late (after two days) in relation to the exposure to this drugs and that there was already severe underdevelopment of the mandibular arches and brain at the time of hemorrhage. Consistent with this, a recent review by Rasmussen and Sadler [2010] concluded that there was not enough epidemiological or experimental data to support the vascular disruption hypothesis for OAVS or microtia. They also emphasized the fact that even malformations caused by genetic alterations occur unilaterally and that other factors that act through nonvascular mechanisms can also cause microtia. In addition, the vascular hypothesis does not explain the abnormalities of OAVS occurring in other non-craniofacial structures (e.g.: kidney and vertebra) or the cumulative evidence showing the frequent bilaterality of this condition. Likewise, this hypothesis cannot adequately explain the bilateral cases of isolated microtia. In addition, the epidemiologic data on the association of OAVS and vascular defects has not been conclusive.

Of further interest is the lack of occurrence of microtia in misoprostol-induced embryopathy. Misoprostol, a synthetic analog of prostaglandin E1, is a vasoconstrictive agent known to cause uterine hypoperfusion. Maternal exposure to misoprostol has been reported for individuals presenting mainly with transverse-limb defects, Moebius sequence and arthrogryposis [da Silva Dal Pizzol et al., 2006; Gonzalez et al., 1993; Orioli and Castilla, 2000; Vargas et al., 2000].

Alternatively, vascular disruptions may simply be an indirect consequence of excessive mesenchymal cell death, perturbed NCC migration or premature NCC differentiation. For example, studies with *Frem2* deficient mice have shown frequent hematomas, yet *Frem2* itself is not expressed in the embryonic vasculature [Timmer et al., 2005]. However, it is expressed in many cell types that contact the vasculature. *Frem2*, and its related proteins, *Fras1* and *Frem1*, have been implicated in the regulation of extracellular matrix structure. Thus, loss of any of these proteins is thought not only to impact tissue morphogenesis but also to increase tissue fragility [Vrontou et al., 2003]. In the uterine environment, the external surface of the developing embryo is constantly in contact with the uterine wall. In this context, increased tissue fragility or reduced cell-specific adhesiveness may increase the embryos susceptibility to physical or mechanical trauma [Vrontou et al., 2003], thus resulting in local vascular disruptions and transient focal tissue ischemia.

Altitude

Castilla et al [1986] reported a five-fold higher prevalence of microtia in Quito, Ecuador (located at 2,850 m or 9,350 ft) compared with countries in low altitudes of South America. The authors proposed that this difference was related to the high altitude. Their analysis did not detect differences in occurrence of microtia among mothers who identified themselves as having Native American ancestry and those that did not. A subsequent study that included Quito and the other two other large high altitude cities of South America, La Paz (Bolivia) and Bogota (Colombia), also revealed a higher prevalence of microtia as well as oral clefts, congenital heart disease and limb defects [Castilla et al., 1999]. The relationship between altitude and microtia is further supported by a recent study using data from vital statistics from Ecuador [González-Andrade et al., 2010], although data on ethnicity in this study was also obtained through self-reporting only. Our literature review failed to identify studies on the prevalence of birth defects in other high-altitude areas such as Tibet.

Intra-uterine growth restriction and increased frequency of preeclampsia and stillbirths are more common in populations living at high altitude than those at low altitude. The uterine artery undergoes remodeling during pregnancy to accommodate the rise in maternal uterine artery blood flow and facilitates delivery of oxygen and nutrients to the feto-placental

circulation. The chronic hypoxia associated with residence at high altitude impairs maternal vascular adaptation to pregnancy by reducing the increase in the uterine artery diameter and rise in its blood flow by about one third. Furthermore, circulating levels of catecholamines and inflammatory cytokines increase during pregnancy in women residing at high altitude [Coussons-Read et al., 2002]. Therefore, high altitude may constrain fetal growth through exposure to low oxygen levels [Zamudio et al., 2006]. Nonetheless, the effects of hypoxia on the developing embryo are not well understood. Evidence from experimental studies suggest that periods of severe hypoxia in the first trimester can cause birth defects, such as transverse limb defects, heart defects, cleft lip and maxillary hypoplasia. However, these studies do not report on anomalies of the ear.

Notably, populations with many generations of residence at high altitude, such as Andean or Tibetan peoples, are relatively protected against this altitude-associated reduction in fetal growth, providing further support for direct biologic altitude effects [Bennett et al., 2008; Julian et al., 2007; Julian et al., 2009]. The mechanisms responsible for the ancestry-associated differences are unclear but could provide important insight into the genetic contributions to microtia. An important confounder, however, is that altitude may constrain agricultural production and thus increase costs of transporting fresh food products. This could feasibly result in maternal nutritional deficiencies that in turn could be the cause for some of the birth defects observed in this population [Cook et al., 2005; Niermeyer et al., 2009; Niermeyer et al., 1995].

A number of genes and respective pathways and environmental factors are required for normal development of the ear. The present challenge is to understand 1) how they integrate to result in the formation of the ear, 2) how their disruption can cause microtia, and 3) how to study new risk factors that may cause microtia.

FUTURE DIRECTIONS

Genetic studies on humans

The prevalence of microtia varies by region. However, even in the higher prevalence regions, and considering the clinical heterogeneity of this condition, the collection of cases requires many years to accomplish a sample size necessary to study this condition. National and international consortiums that include data from birth defect registries and/or craniofacial centers could facilitate prospective, standardized data collection for individuals with microtia. Such data would increase the likelihood of success for larger studies and thus, advance the knowledge of the etiology of microtia.

Recent advances offer multiple methodologies to study the genetics of microtia. The four most common methods include genome-wide association studies (GWAS), exome sequencing, linkage studies in large families, and copy number variation investigations. The success of such investigations requires high-quality phenotypic data.

Given that the prevalence of microtia appears to be higher in some ethnic groups, we would expect at least in these groups, that the genetic variants associated with microtia are common (i.e., present in more than 5% of the population) and, therefore, GWAS could be a suitable and cost-effective approach. Although GWAS studies are typically not feasible for studies of birth defects given the need for large sample sizes, a successful GWAS study with 111 cases was recently performed in oral clefting [Grant et al., 2009]. New GWAS arrays also allow for the clarification of ethnicity through the identification of ancestry informative markers and the identification of copy number variations providing information regarding cytogenetic diagnoses.

Exome sequencing offers promise as a technique to study microtia, particularly in isolated cases from ethnic groups that apparently have a lower risk for microtia. In such cases the genetic effects seem to be rare and therefore they could potentially represent sporadic variants. Exome sequencing can identify coding variants specific to each individual studied and some functional annotation can usually be ascribed to the findings. In contrast, functional annotation in GWAS studies usually has to be inferred via linkage disequilibrium as assessed variants are not necessarily functional variants. Another advantage of exome sequencing is the option to study fewer cases (such as case-parent trios) and to identify genetic variation, although this technology has not yet been proven to be effective for complex diseases. Exome sequencing techniques are limited by the fact that they do not identify functional noncoding, nor structural, mutations; however, the recent development of analysis of copy number variation data derived from exome sequencing could partially overcome this limitation. Identification of mutations in segmentally duplicated regions of the genome with short read sequencing also remains challenging in exome sequencing.

Animal models

There are likely many murine models with ear abnormalities not yet described in the literature. Identification of microtia in animal models can be challenging for the following reasons: 1) mild types of microtia can easily go unnoticed to an unfamiliar handler; 2) experiments that are focused on other phenotypes may not report abnormalities of the ear; 3) even if noticed, the likelihood of having it reported and/or published is low. This is supported by our own investigations of numerous existing mutant mouse lines that have shown many striking yet previously unreported craniofacial malformations including microtia (T. Cox, unpublished data). In addition, auditory canal and tympanic membrane abnormalities are more difficult to identify in mouse models than pinna defects.

A concerted effort assessing the large repositories of spontaneous, chemically-induced and gene targeted mouse lines such as the Jackson Laboratories (<http://www.informatics.jax.org>) may ultimately uncover many new and important genes involved in external ear development and hence candidates for microtia in humans.

CONCLUSIONS

The genetic and cellular mechanisms underlying normal morphogenesis of the external ear is not completely understood. Further insight into the mechanisms of normal ear development will contribute to an understanding of abnormal ear development that results in microtia and other ear abnormalities. Identification and characterization of the primary and secondary factors that lead to microtia on the other hand will be important for the delineation of the molecular pathways involved in craniofacial development. In addition, such studies will likely open new strategies for treatment for individuals with microtia. In conjunction with well designed clinical research, continued application of novel technologies and models is essential to fully understand the pathogenesis of isolated microtia and the exact role that individual genes play in the development of the external ear.

Acknowledgments

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Figure 1. Photographs of individuals with different types of microtia (*compose* of 10 photos). Top left shows a typical ear.

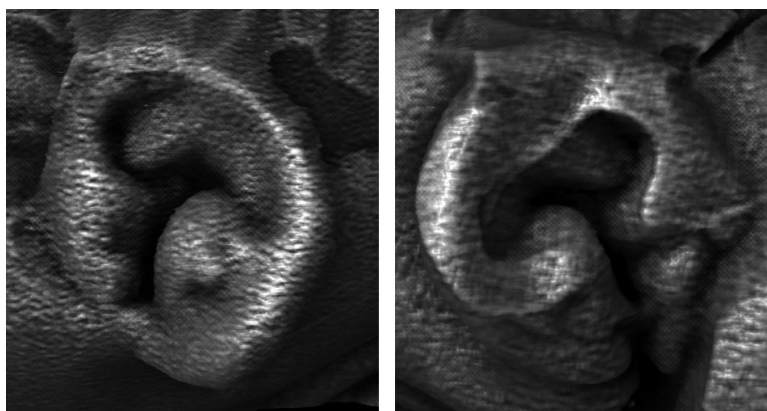


Figure 2. External ear morphology in a 122d anencephalic fetus. 3D rendered image of a microcomputed tomography (microCT) scan showing external ear morphology. Note the microtic appearance of the right ear (right image) compared to the normal left ear (left image).

Table I

Prevalence (per 10,000) of microtia reported in the literature from 1960-2010.

Study	Age of Ascertainment	Prevalence Microtia-Anotia	Prevalence Anotia	Prevalence Microtia	Types of microtia included (a)	Authors	Year
Population-based							
Central East France	LB+SB	0.8	0.4	0.4	I-IV	Harris et al (c)	1996
California (USA)	LB+SB	2.0	0.2	1.8	I-IV	Harris et al (c)	1996
Sweden	LB+SB	2.4	0.2	2.1	I-IV	Harris et al (c)	1996
California (USA)	LB+SB	2.2	nr	nr	nr	Shaw et al (c)	2004
Hawaii (USA)	LB+SB+ETOP	3.8	0.3	3.5	II-IV	Forrester et al	2005
Finland	LB+SB	4.3	0.2	4.1	I-IV	Suurtala et al	2007
Texas (USA)	LB+SB+ETOP	2.8	nr	nr	nr	Husain et al	2008
Texas (USA)	LB+SB+ETOP	2.9	0.2	2.7	II-IV	Canfield et al	2009
Hospital-based							
Navajo (USA)	> 21 years old	9.7	nr	nr	nr	Jaffe et al	1967
New Mexico (USA)	Any age	1.3	nr	nr	I-IV	Aase et al	1977
Navajo (USA)	4-14 years old	12.0	nr	nr	nr	Nelson et al	1984
South America (b)	LB	3.2	nr	nr	I-IV	Castilla et al	1986
Italy	LB+SB	1.5	0.3	1.2	I-IV	Mastroiacovo et al	1995
China	LB+SB	1.4	nr	nr	nr	Zhu et al	2000
Chile	LB+SB	8.8	0.5	8.3	I-IV	Nazer et al	2006

LB=Livebirth, SB=Stillbirth, ETOP=Elective termination of pregnancy

nr= not reported

(a) According to Marx Classification

(b) Argentina, Brazil, Chile, Ecuador, Peru, Uruguay, and Venezuela

(c) Excluded cases with known chromosomal anomalies

Table II

Human disorders with microtia (except Chromosomopathies)

Syndrome	Microtia (%) ^(a)	Gene(s) identified
Bixler (Hypertelorism-microtia-clefting)	100	--
Bosley-Salih-Alorainy	33	<i>HOXA1</i>
Branchiooculofacial (BOF)	20	<i>TFAP2A</i>
Branchiootic (BO)	80-90	<i>EYA1</i> <i>SIX1</i>
Branchiootorenal (BOR)	30-60	<i>EYA1</i> <i>SIX5</i>
CHARGE	Reported	<i>CHD7</i> <i>SEMA3E</i>
Congenital deafness, inner ear agenesis, microtia, microdontia	100	<i>FGF3</i> ^(b)
Craniofacial microsomia (CFM)	65	--
Fraser	Reported	<i>FRAS1</i> <i>FREM2</i>
Kabuki	80	<i>MLL2</i>
Klippel-Feil	Reported	<i>GDF6</i>
Lacrimoauriculodentodigital	20	<i>FGFR2</i> <i>FGFR3</i> <i>FGF10</i>
Mandibulofacial dysostosis	100	<i>HOXD</i>
Meier-Gorlin (Ear-patella-short stature)	100	<i>ORC1</i> <i>ORC4</i> <i>ORC6</i> <i>CDT1</i> <i>CDC6</i>
Microtia, hearing impairment, and cleft palate	100	<i>HOXA2</i>
Miller	100	<i>DHODH</i>
Nager	80	--
Oculo-auricular	100	<i>HMX1</i>
Pallister Hall	Reported	<i>GLI3</i>
Townes Brocks	20	<i>SALL1</i>
Treacher-Collins	60-80	<i>TCOF1</i>
Wildervanck (Cervicooculoacoustic)	Reported	--

^(a) Gorlin et al., 2001^(b) Hemizigosity: only inner ear anomalies

Table III

Risk factors for microtia reported in the literature

Risk factors	Author and year of publication ^(a)
General	
Male sex	all authors, except Zhu et al., 2000
Low birthweight	Castilla and Orioli, 1986; Forrester and Merz, 2005; Mastroiacovo et al., 1995
First parity	Mastroiacovo et al., 1995
High parity	Castilla and Orioli, 1986; Harris et al., 1996; Mastroiacovo et al., 1995
Multiple births	Forrester and Merz, 2005; Shaw et al., 2004
Maternal acute illnesses	Castilla and Orioli, 1986; Okajima et al., 1996; Zhang et al., 2009 ^(b)
Maternal insulin dependent diabetes	Correa et al., 2008; Mastroiacovo et al., 1995; Stevenson, 2006
Maternal use of medications	Castilla and Orioli, 1986; Zhang et al., 2009 ^(c)
Advanced paternal age	Castilla and Orioli, 1986
Advanced maternal age	Forrester and Merz, 2005; Harris et al., 1996
Low maternal education	Harris et al., 1996; Shaw et al., 2004; Zhang et al., 2009
Maternal exposure to altitude	Castilla et al., 1999
Maternal residence in an urban area	Zhu et al., 2000
Maternal residence in a rural area	Zhang et al., 2009 ^(d)
Maternal exposure to air pollution	Zhang et al., 2009 ^(e)
Race/Ethnicity	
Native ethnicity	Aase and Tegtmeier, 1977; Jaffe, 1969
Hispanic ethnicity	Harris et al., 1996; Husain et al., 2008; Shaw et al., 2004
Ecuadorian	Castilla and Orioli, 1986
Chilean	Nazer et al., 2006
Asian, Philippine, Pacific Islander	Forrester and Merz, 2005
Teratogens	
Retinoic acid	Carey et al., 2006; Lammer et al., 1985; Stern et al., 1984
Thalidomide	Carey et al., 2006
Alcohol	Carey et al., 2006
Mycophenolate mofetil	Anderka et al., 2009; Merlob et al., 2009; Perez-Aytes et al., 2008

^(a) Case-control studies: Castilla and Orioli, 1986; Mastroicovo et al, 1995; Zhang et al, 2009.

^(b) No specific maternal condition identified

^(c) No specific drug identified

^(d) Controls were not defined

^(e) Controls and data on pollution were not defined

Table IV

Classification systems for microtia

<i>Marx, 1926</i>	<i>Tanzer, 1978</i>
Grade I. Abnormal auricle with all identifiable landmarks. Grade II. Abnormal auricle without some identifiable landmarks. Grade III. Very small auricular tag or anotia. Rogers (1977) proposed a fourth grade classification: Grade IV. Anotia.	Type 1. Anotia. Type 2. Completely hypoplastic ear (microtia). a. With atresia of the external auditory canal. b. Without atresia of the external auditory canal. Type 3. Hypoplasia of the middle third of the auricle. Type 4. Hypoplasia of the superior third of the auricle. a. Constricted (cup and lop) ear. b. Cryptoptia. c. Hypoplasia of entire superior third. Type 5. Prominent ear
<i>Weerda, 1988</i>	<i>Hunter et al, 2009</i>
First degree dysplasia. Most structures of a normal auricle are recognizable (minor deformities). A. Macrotia E. Small deformities B. Protruding ears F. Colobomata C. Cryptoptia G. Lobule deformities D. Absence of upper helix H. Cup ear deformities Second degree dysplasia. Some structures of a normal auricle are recognizable. A. Cup ear deformity type III B. Mini ear Third degree dysplasia. None of the structures of a normal auricle are recognizable. A. Unilateral B. Bilateral C. Anotia (Peanut ears are included in this group)	Microtia, First Degree. Presence of all the normal ear components and the median longitudinal length more than 2 SD below the mean. Microtia, Second Degree. Median longitudinal length of the ear more than 2 SD below the mean in the presence of some, but not all, parts of the normal ear. Microtia, Third Degree. Presence of some auricular structures, but none of these structures conforms to recognized ear components. Anotia. Complete absence of the ear.

Table V

Summary of genes associated with microtia in mice and comparison with ear phenotypes in humans.

Gene	Mouse ear phenotype	Human ear phenotype
<i>Bmp5</i>	Small outer ear/ Normal EAM	nr
<i>Chuk (Ikka)</i>	Anotia	nr (MIM:613630- no ear anomalies in the two cases with <i>IKKA</i> mutations)
<i>Cyp26b1</i>	Small outer ear	nr
<i>Dlx5/Dlx6</i>	Absent outer ear Severely affected middle ear	nr
<i>Edn1</i>	Very small outer ear/ Absent EAM/ Absent middle ear	nr
<i>Ednra</i>	Small outer ear	nr
<i>Eya1</i>	Anotia	<i>EYA1</i> : 30% with outer ear anomalies (overfolded/ deformed) ^(a)
<i>Fgf8</i>	Small outer ear/Middle ear anomalies	nr
<i>Fgf10</i>	Small outer ear	nr
<i>Frem2</i>	Anotia	nr
<i>Gsc</i>	Slightly reduced outer ear/ EAM absent/Inner ear anomalies	nr
<i>Hfm locus</i>	Small outer ear/Anotia/ Absent EAM/ Middle ear anomalies	Correlated gene not identified in humans
<i>Hoxa1</i>	Small outer ear/ Middle and inner ear anomalies	Deformed ears (present in few cases)
<i>Hoxa1/Hoxb1</i>	Anotia (with no residual auricle)	nr
<i>Hoxa2</i>	“Anotia” (poorly defined protuberance)	Microtia I and II ^(a)
<i>Hox2.2</i>	Small outer ear	nr (Humans: <i>HOXB6</i>)
<i>Irf6</i>	Anotia	nr
<i>Pax8</i>	Anotia/ Small outer ear/ EAM stenotic or atretic/ Middle and inner ear anomalies	nr
<i>Prrx1/Prrx2</i>	Small outer ear/ Middle and inner ear anomalies	nr (Humans: <i>PMX1</i>)
<i>Prkra</i>	Small outer ear / Middle ear anomalies	nr
<i>Rar</i>	Small outer ear/ Anotia	nr
<i>Sall1</i>	No affected ears	Microtia types I-II ^(a) /Preauricular tags
<i>Six1/Six4</i>	Anotia	nr
<i>Tbx1</i>	Small outer ear/ Anotia / Middle and inner ear anomalies	Microtia I ^(a) (uncommon)
<i>Tcfap2a</i>	Anotia	nr
<i>Tcof1</i>	Cup-shaped ears/ Middle ear anomalies	Microtia and ear canal atresia.
<i>Wnt5a</i>	Small outer ear	nr

nr: not reported

^(a) According to Marx Classification